Sonu Kumar

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Professional Summary

DevOps Engineer with 1 year of hands-on internship experience building streamlined cloud infrastructure, end-to-end CI/CD pipelines, and containerized deployments on AWS. Proficient in Kubernetes, Docker, Terraform, Ansible, and Jenkins, with a strong focus on DevSecOps practices, zero-downtime release strategies, and system observability. Committed to accelerating delivery cycles, eliminating manual bottlenecks, and engineering scalable, secure cloud-native solutions.

Technical Skills

Cloud – AWS: EC2, S3, EBS, VPC, ELB, Lambda, DynamoDB, Fargate, ECS, EKS, ECR, SNS, SQS, CloudFront, CloudWatch, API Gateway, CloudFormation, Auto Scaling, NAT Gateway, WAF, Network ACLs, Systems Manager, AWS Global Accelerator

CI/CD & DevSecOps: Jenkins, Azure DevOps, GitHub Actions, SonarQube, Maven, Nexus, Trivy, OWASP Dependency-Check, Docker Hub

Infrastructure as Code: Terraform (modules, remote state, state locking), Ansible (playbooks, roles), CloudFormation, YAML

Containers & Orchestration: Docker, Kubernetes – Deployments, Services, Pods, Namespaces, ConfigMaps, Secrets, Ingress, RBAC, CronJobs, Node Port, Load Balancer

Monitoring & Observability: Prometheus, Grafana, ELK Stack (Elasticsearch, Logstash, Kibana), CloudWatch, Application Performance Monitoring (APM)

Version Control: Git, GitHub, GitLab – GitFlow branching, pull requests, collaborative code review

Scripting & Programming: Bash, Python, PowerShell, Groovy, Shell scripting

Networking & Security: Linux (Ubuntu, CentOS), TCP/IP, IP subnetting, IAM, SSH, SSL/TLS, Security Groups, Network Firewall, Vulnerability Assessment

Professional Experience

JSpiders – Training & Development Center — **March 2025 – March 2026**
DevOps Engineer Intern — *Marathahalli, Bengaluru, Karnataka*

- Designed and maintained CI/CD pipelines in Jenkins with Git webhook triggers, systematizing the build, test, security scan, and multi-environment deployment lifecycle to cut release cycle time substantially.
- Provisioned scalable AWS infrastructure – including EC2, VPC, S3, ELB, and IAM roles – via Terraform, producing consistent, version-controlled platforms across development and production.
- Containerized microservices with Docker and deployed them on Kubernetes, configuring Ingress controllers, ConfigMaps, Secrets, and RBAC policies to govern access control and traffic management.
- Strengthened pipeline security by embedding SonarQube quality gates, Trivy container scans, and OWASP Dependency-Check into the CI workflow, intercepting critical vulnerabilities before every release.
- Established centralized observability through Prometheus and Grafana dashboards alongside the ELK Stack, powering real-time alerting, distributed log aggregation, and performance tracking.
- Administered Linux servers (Ubuntu, CentOS), performing patch management, file system tuning, security hardening, and root-cause analysis to maintain high system availability.
- Developed reusable Ansible playbooks to standardize server configuration and remove repetitive provisioning tasks, reducing setup time throughout the team.
- Managed all codebase changes in GitHub, following GitFlow branching conventions and conducting peer reviews to uphold code quality and traceability.

Projects

Blue-Green Deployment on AWS with Kubernetes (kOps) | *AWS EC2, S3, ELB, kOps, Docker, Nginx, Tomcat, Terraform*F

- Stood up a production-grade Kubernetes cluster on AWS via kOps, with Amazon S3 as the remote state store to support team-wide, high-availability cluster management.
- Architected a Blue-Green deployment model with dual Kubernetes Deployments behind a single Elastic Load Balancer, facilitating instantaneous traffic cutover between versions with zero downtime.
- Leveraged Kubernetes Service selectors to shift live traffic between environments without Pod redeployment, yielding sub-second failover capability and continuous service availability.
- Configured ELB health checks and security group rules to guarantee reliable request routing, automatic fault detection, and self-healing behavior across worker nodes.
- Connected CloudWatch metrics with Prometheus alerting rules to track cluster health, node performance, and uptime, supporting proactive incident response.

DevSecOps CI/CD Pipeline – Zomato Clone Application | *Jenkins, Docker, SonarQube, Trivy, OWASP, AWS EC2, Ansible,*

- Built a fully programmatic Jenkins pipeline spanning Maven build, static code analysis, OWASP dependency inspection, container image vulnerability scanning, publication to Docker Hub, and EC2 deployment.
- Enforced SonarQube Quality Gates as a mandatory gating stage, blocking any release that fell below minimum code coverage thresholds or security benchmarks.
- Implemented instant reversion logic in Jenkins to restore the last known-good container image upon deployment failure, reducing mean time to recovery (MTTR) and protecting service continuity.
- Eliminated over 90% of manual deployment effort through end-to-end pipeline orchestration, delivering a containerized workload from code commit to a live instance on AWS EC2.

Infrastructure as Code (IaC) on AWS with Terraform | *AWS EC2, S3, VPC, Terraform, Ansible, GitHub* **Dec 2025**

- Authored modular Terraform configurations for VPC, subnets, EC2 instances, IAM roles, and S3 buckets, applying DRY principles to maximize reusability and apply uniform standards across all systems.
- Secured remote state in AWS S3 with DynamoDB-based locking, preventing concurrent modification conflicts and ensuring safe, coordinated provisioning operations.
- Connected the Terraform codebase to GitHub with pull request workflows, enabling peer-reviewed changes, full audit trails, and point-in-time state recovery.

Automated Backup and Disaster Recovery System | *Bash, Python, Ansible, AWS S3, Cron* **Dec 2025**

- Scripted daily database and file system backup routines in Bash and Python, compressing and encrypting archives before transferring them to AWS S3 for durable, offsite retention.
- Scheduled jobs via Cron with configurable retention and rotation policies, optimizing storage costs while meeting defined recovery point objectives (RPO).
- Orchestrated cross-server backup verification and restoration drills through Ansible playbooks, validating data integrity and confirming disaster recovery readiness on a recurring basis.

Education

Gnanamani College of Technology – Anna University **May 2021 – May 2025**
Bachelor of Engineering, Computer Science and Engineering *Namakkal, Tamil Nadu*

- CGPA: 7.48 / 10

Ayodhya Prasad Higher Secondary School
Senior Secondary (Class XII) *Ayodhya Ganj, Katihar*

AP High School
Secondary (Class X) *Ayodhya Ganj, Katihar*

Certifications

DevOps Engineer Training Certificate – JSpiders Software Training *Linux, Docker, Kubernetes, AWS, CI/CD, Jenkins*
Networking Essentials – Cisco Networking Academy *Network fundamentals, routing & switching*
Advanced Kubernetes Bootcamp – Udemy *Cluster management, production-grade deployments*
DevOps Tools for Beginners – Udemy *CI/CD pipelines, DevOps toolchain fundamentals*
Web Development Certificate – GTL Software Pvt. Ltd. *HTML, CSS, JavaScript, full-stack basics*